
title: "Microsoft Innovate 2026 — Round 1
Idea Submission" subtitle: "Problem
Statement #26 — Catch the Attack the
Signatures Miss"

Microsoft Innovate 2026 — Round 1 Idea Submission

Team Name: [Enter your team name] **Team ID:** [As given at registration] **Problem Statement Title:** #26 — Catch the Attack the Signatures Miss **Theme / Bucket:** Cybersecurity — Security Operations **Team Leader:** [Name, Enrollment No., Contact Email] **Submission deadline:** [DD Month 2026]

Team Details

S. No.	Full Name	Enrollment No.	Programme & Year	Email	Role in Team
1	[Name]	[Enrollment no.]	[e.g. B.Tech CSE, 2nd Year]	[email]	Team Leader
2	[Name]	[Enrollment no.]	[Programme, Year]	[email]	[e.g. Developer, Designer]
3	[Name]	[Enrollment no.]	[Programme, Year]	[email]	[e.g. Developer, Designer]

Teams of 2 to 6 members. Delete unused rows. Faculty Mentor (optional): [Name, Department, Email]

Problem Statement

What is the problem?

- Signature-based IDS only match known attack fingerprints.
- Independent tests: Snort misses ~83% of zero-day attacks.
- SOC teams already see 46–53% of alerts as false positives.

Who faces it?

- Tier-1 SOC analysts at any org running a signature IDS/SIEM — banks, SaaS firms, MSSPs.
- Every shift, every day — alert queues refill continuously, not as a one-off incident.

Why do current options fall short?

- Snort / Suricata (signature IDS) and SIEM correlation rules — both need a known pattern to fire.
- Gap: no lightweight, explainable ML layer that flags anomalous traffic analysts can actually trust and act on.
- Novel attack variants keep slipping through the pattern-matching net, sometimes undetected for months.

Proposed Solution

Our idea in one line

NetSentinel is an ML classifier that flags what signature IDS misses — ranked SOC alerts, never an auto-block.

How it solves the problem

- Random Forest classifier tags each network flow normal vs. attack, plus an attack-type label (DoS, brute force, port scan...).
- Every alert ships with a confidence score and top contributing features, so an analyst can verify it in seconds.
- Nothing is auto-blocked — flagged traffic lands in a SOC queue for acknowledge / escalate, keeping a human in the loop.

What makes it different

- Honest evaluation, not accuracy-only spin — precision, recall, false-positive rate and AUC reported per attack class.
- Real Azure usage: Azure Machine Learning for training and model registry, Azure OpenAI to draft a plain-English brief per incident.
- Every alert is explainable — SHAP feature attribution shows analysts why, not just what.

(Add a rough sketch/mockup of the SOC alert queue here if you have one.)

Technical Approach

Planned tech stack

Layer	Choice
Frontend	Streamlit / React
Backend	FastAPI (Python)
Database	SQLite
AI / APIs	scikit-learn, SHAP

Layer	Choice
Cloud	Azure ML + OpenAI
Other tools	CICFlowMeter, SMOTE

How the system will work

Network traffic flows (CICIDS2017 / live PCAP) → SOC alert queue dashboard → **Random Forest classifier + SHAP** → Ranked alerts – analyst acknowledges / escalates

What we have started working on

- CICIDS2017 downloaded and profiled; cleaning + feature-extraction pipeline built.
- First-pass Random Forest baseline trained, with precision / recall / AUC logged to confirm feasibility.

Impact and Next Steps

Expected impact

- SOC analysts get triaged, explainable alerts instead of a raw firehose of noise.
- Target: cut false-positive rate well below the 46–53% industry baseline while keeping attack recall high.

Feasibility

- CICIDS2017 / NSL-KDD / UNSW-NB15 are free and labelled; Random Forest trains in minutes on a laptop.
- Working Python, pandas and scikit-learn experience — a data pipeline plus one well-understood model.

Plan for the next round

1. Clean the data and train an honest baseline — precision / recall / F1 / AUC reported per attack class.
2. Add the SOC alert dashboard with acknowledge / escalate actions and SHAP explanations.
3. Demo: replay traffic containing a planted, signature-evading attack and show it correctly flagged live.

References and data sources

- CIC — CICIDS2017 Intrusion Detection Dataset (unb.ca)
- Sabottke et al. — Signature IDS vs. Zero-Day Attacks (IEEE)
- Microsoft/Omdia, State of the SOC — 46% of alerts are false positives
- SANS 2025 Survey — false positives are SOC teams' #1 challenge

